

Cloud Platform Architecture over Virtualized Data Centers

Module 3.1

Module 3.1 Topics

	3	Cloud Infrastructure and Platforms in Industry	10	CO2
	3.1	Data-Center Design-Warehouse-Scale Data-Center, Modular Data Center. Interconnection Networks-Fat-Tree Interconnection Topology for Warehouse Scale Data Centers, Server-Centric BCube Network for a Modular Data Center.Data-Center Management Issues		
	3.2	Amazon Web Services – Compute Services, Storage Services, Communication Services, Additional Services		
	3.3	Google Cloud Platform, Google AppEngine: Architecture and Core concepts; Application Life Cycle		

Module 3.1 Topics

Book-

- Title-Distributed and Cloud Computing
- Authors-Kai Hwang, Geoffrey C.Fox, Jack . Dongarra

DATA-CENTER

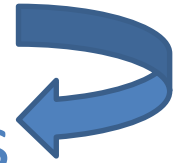
- A data center is often built with a large number of servers through a huge interconnection network.

DATA-CENTER DESIGN AND INTERCONNECTION NETWORKS

- A data center is often built with a large number of servers through a huge interconnection network.
- We will study the design of
 - large-scale data centers and
 - small modular data centers that can be housed in a 40-ft truck container.
 - interconnection of modular data centers and their management issues and solutions.

DATA-CENTER DESIGN AND INTERCONNECTION NETWORKS

- Warehouse-Scale Data-Center Design
- Data-Center Interconnection Networks
- Modular Data Center in Shipping Containers
- Interconnection of Modular Data Centers
- Data-Center Management Issues .



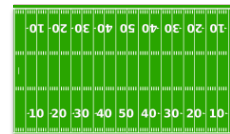
Warehouse-Scale Data-Center Design

- Dennis Gannon claims: “The cloud is built on massive data centers” .



Warehouse-Scale Data-Center Design

- The Figure shows a data center that is **as large as a shopping mall (11 times the size of a football field) under one roof.**
- Such a data center can house **400,000 to 1 million servers.**
- The data centers are built economics of scale—meaning lower unit cost for larger data centers.
- A small datacenter could have 1000 servers.



**This data center is
11.5 times
the size of a football field**

Economics of scale

- The approximate monthly cost to operate a huge 400-server data center is estimated by
 - network cost \$13/Mbps;
 - storage cost \$0.4/GB; and
 - administration costs.
- These unit costs compared to a 1,000-server data center.
 - The network cost of a small data center is about seven times greater
 - The storage cost is 5.7 times greater.

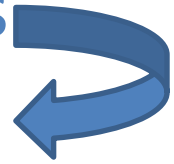
Economics of scale

Inference-

- The larger the data center, the lower the operational cost.

Warehouse-Scale Data-Center Design

- Data-Center Construction Requirements
- Cooling System of a Data-Center Room



Data-Center Construction Requirements

- Most data centers are built with commercially available components.
- An off-the-shelf server consists of :
 - a number of processor sockets,
 - each with a multicore CPU and
 - its internal cache hierarchy,
 - local shared and coherent DRAM,
 - a number of directly attached disk drives.

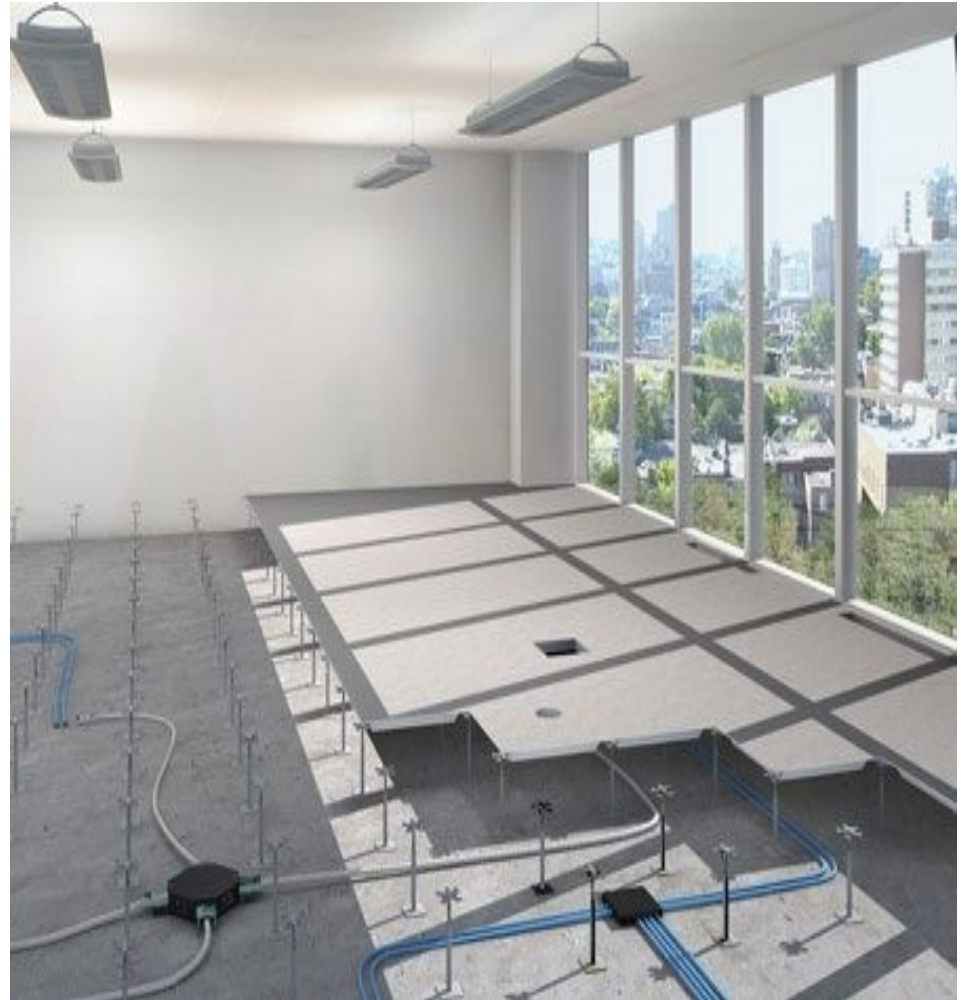
Off the shelf-not designed or made to order but taken from existing stock or supplies.

Data-Center Construction Requirements

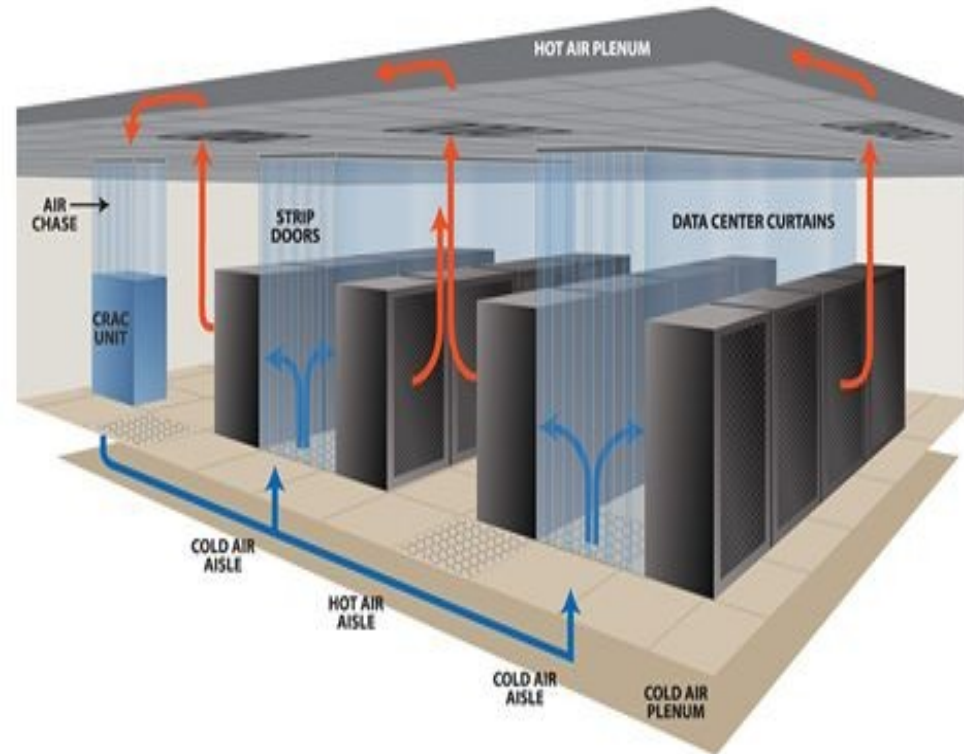
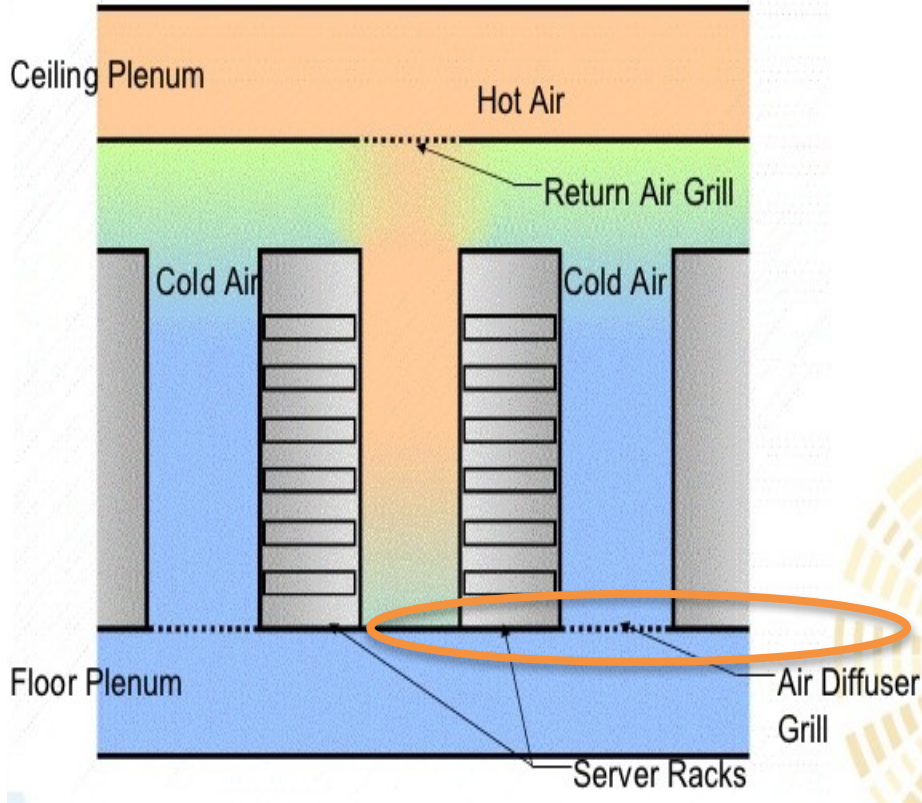
- The DRAM and disk resources within the rack are accessible
 - through first-level rack switches and
 - all resources in all racks are accessible via a cluster-level switch
- Consider a data center
 - built with 2,000 servers,
 - each with 8 GB of DRAM and
 - four 1 TB disk drives.
 - Each group of 40 servers is connected through a 1 Gbps link to a rack-level switch
 - Each Rack switch has an additional eight 1 Gbps ports used for connecting the rack to the cluster-level switch.

Cooling System of a Data-Center Room

- The data-center room has raised floors for hiding cables, power lines, and cooling supplies.
- The raised floor has a steel grid resting on stanchions about 2–4 ft above the concrete floor.
- The under-floor area is often used to route power cables to racks, but its primary use is to distribute cool air to the server rack.

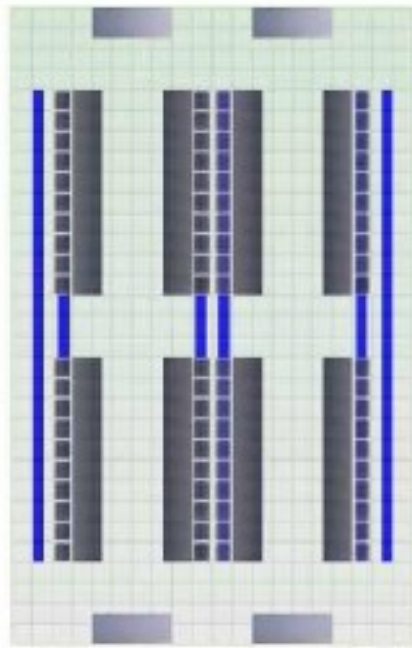


Cooling System of a Data-Center Room

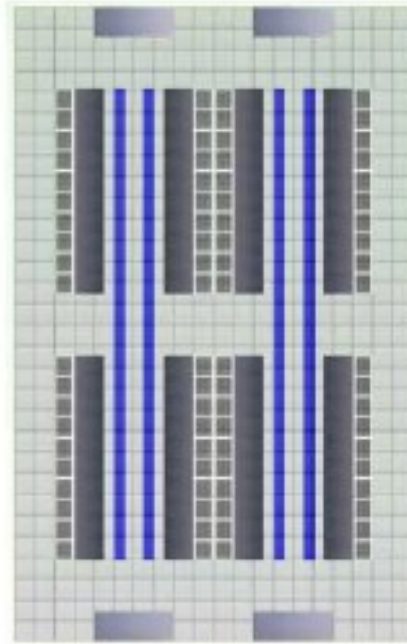


- The CRAC (computer room air conditioning) unit pressurizes the raised floor plenum by blowing cold air into the plenum.
- The cold air escapes from the plenum through perforated tiles that are placed in front of server racks.

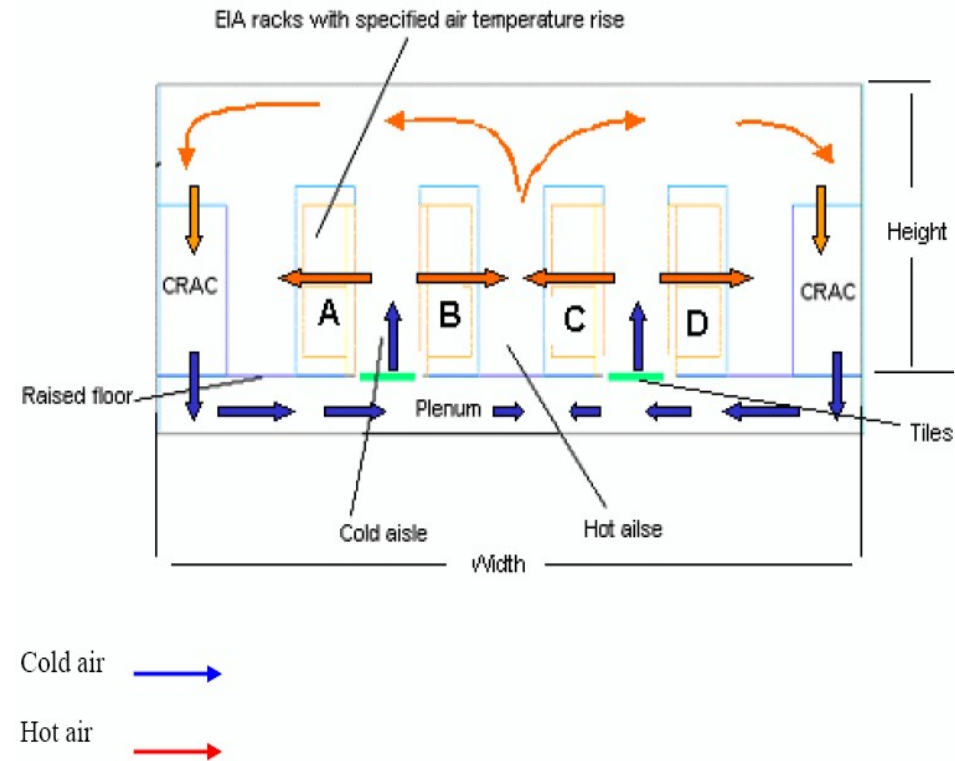
Cooling System of a Data-Center Room



(a) Cold Aisle Cable (CAC)



(b) Hot Aisle Cable (HAC)



- Racks are arranged in long aisles that alternate between cold aisles and hot aisles to avoid mixing hot and cold air.

Cooling System of a Data-Center Room

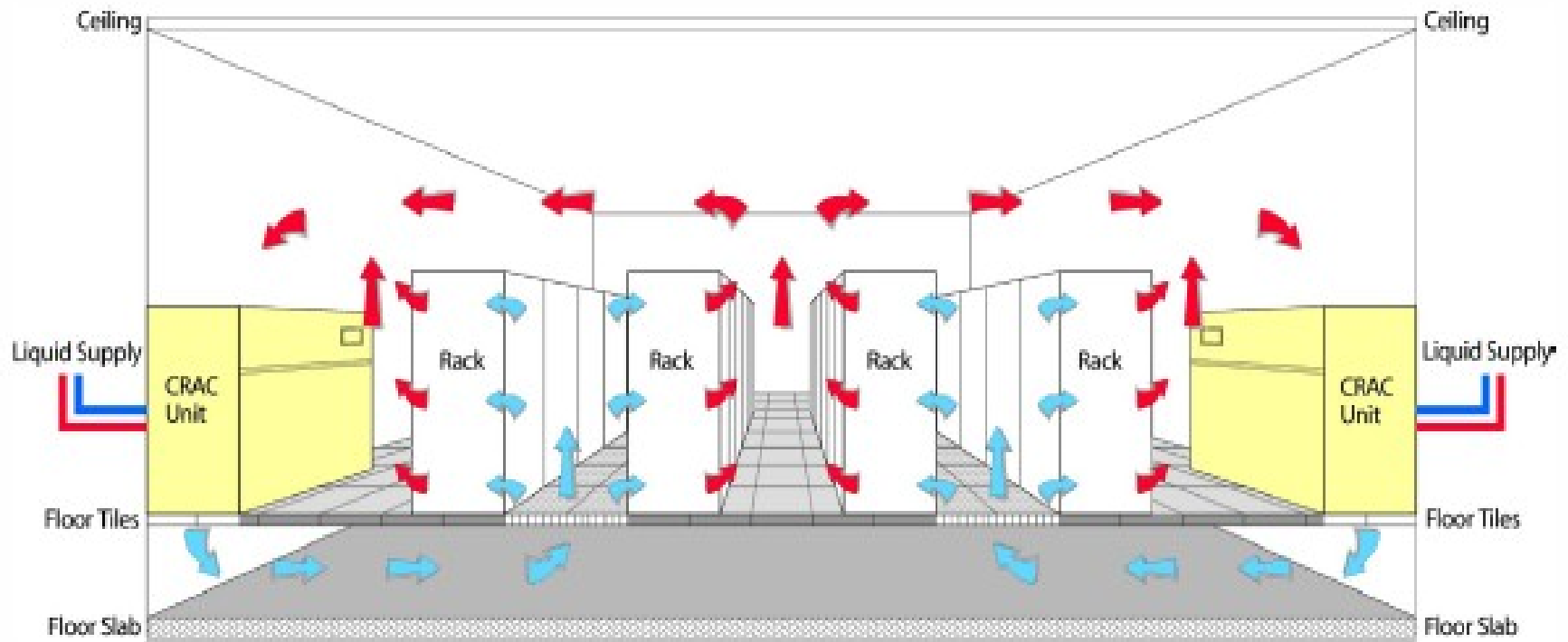


FIGURE 4.2: Datacenter raised floor with hot-cold aisle setup (image courtesy of DLB Associates [23]).

- The Figure shows the layout and cooling facility of a warehouse in a data center.

Cooling System of a Data-Center Room

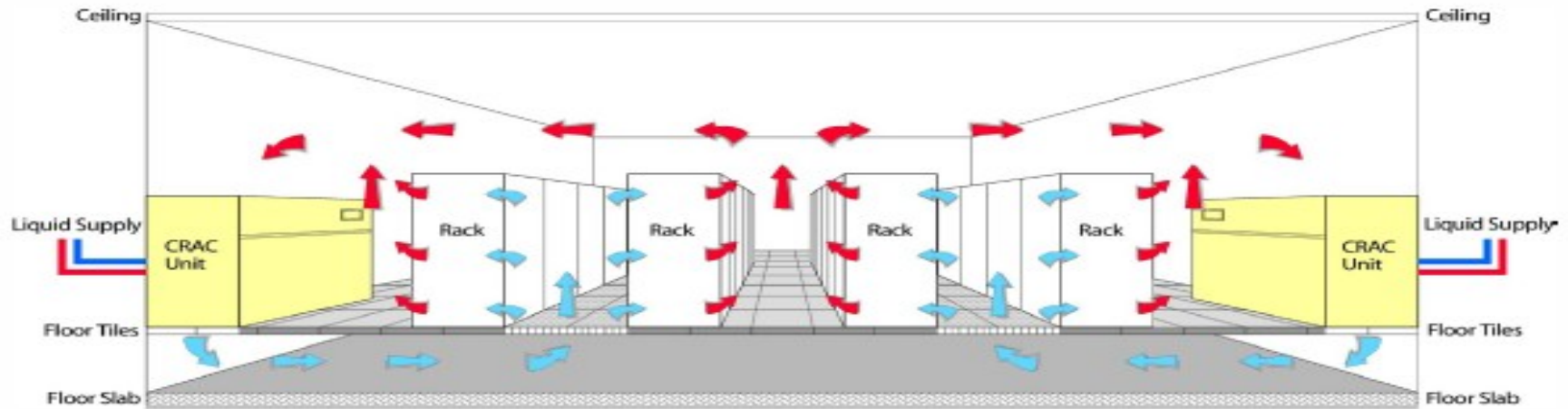
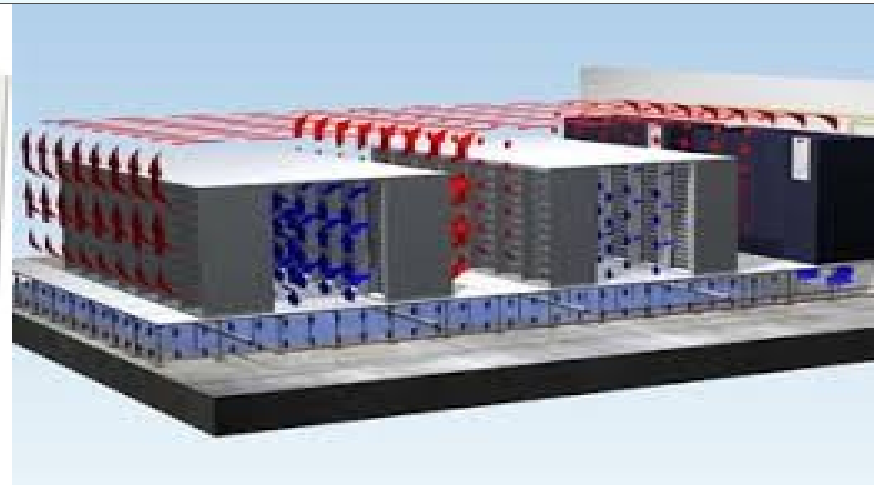
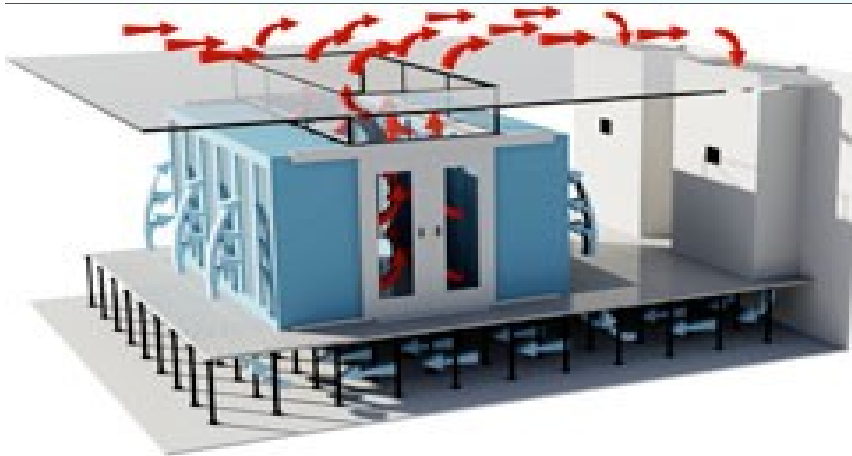
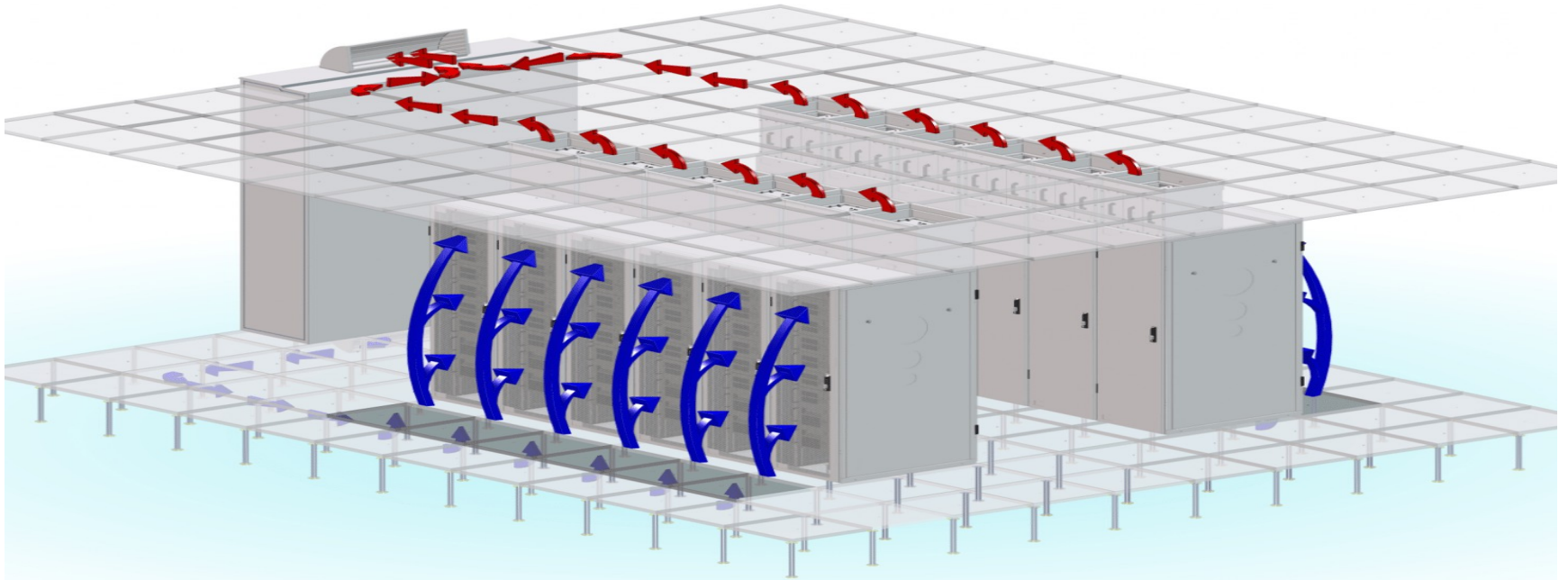


FIGURE 4.2: Datacenter raised floor with hot-cold aisle setup (image courtesy of DLB Associates [23]).

- The hot air produced by the servers circulates back to the intakes of the CRAC units that cool it and then exhaust the cool air into the raised floor plenum again.
- Typically, the incoming coolant is at 12–14°C and the warm coolant returns to a chiller.

Cooling System of a Data-Center Room



Video Links

- Hot Aisle/Cool Aisle
- <https://youtu.be/yeqfuee1Vkc>

Skanska Data Center Construction Simulation

- https://youtu.be/Pt0_cf5IJJc

Under the sea, Microsoft tests a datacenter

- Microsoft is leveraging technology from submarines and working with pioneers in marine energy for the second phase of its develop self-sufficient underwater datacenters that can deliver lightning-quick cloud services to coastal cities.
- An experimental, shipping-container-size prototype is processing workloads on the seafloor near Scotland's Orkney Islands.



Data-Center Interconnection Networks

- A critical core design of a data center is the **interconnection network among all servers in the data center cluster**.
- This network design must meet five special requirements:
 - low latency
 - high bandwidth,
 - low cost
 - message-passing interface (MPI) communication support
 - fault tolerance

Data-Center Interconnection Networks

Specific design considerations are

- Application Traffic Support
- Network Expandability
- Fault Tolerance and Graceful Degradation
- Switch-centric Data-Center Design

Application Traffic Support

- The network topology should support all MPI(Message Passing Interface) communication patterns.
- Both point-to-point and collective MPI communications must be supported.
- The network should have high bisection bandwidth to meet this requirement.

Application Traffic Support

- For example, one-to-many communications are used for supporting distributed file access.
- One can use one or a few servers as metadata master servers
 - which need to communicate with slave server nodes in the cluster.
- To support The MapReduce programming Paradigm, the network must be designed to perform the map and reduce functions at a high speed.
- The underlying network structure should support various network traffic patterns demanded by user applications

Network Expandability

- The interconnection network should be expandable.
 - With thousands or even hundreds of thousands of server nodes,
 - the cluster network interconnection should be allowed to expand once more servers are added to the data center.
- The network topology should be restructured while facing such expected growth in the future.

Network Expandability

- The fat-tree and crossbar networks could be implemented with low-cost Ethernet switches.
- However, the design could be very challenging when the number of servers increases sharply.

A Fat-Tree Interconnection Network for Data Centers

- The fat-tree topology is applied to interconnect the server nodes.

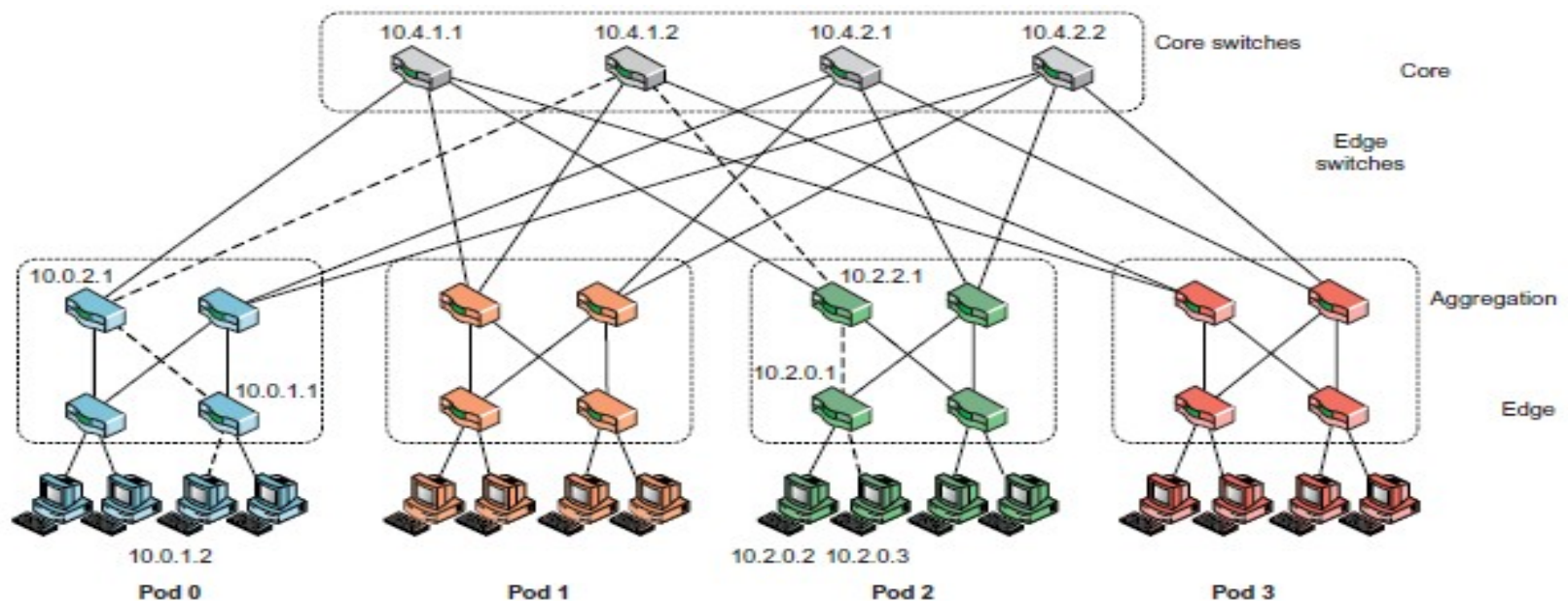
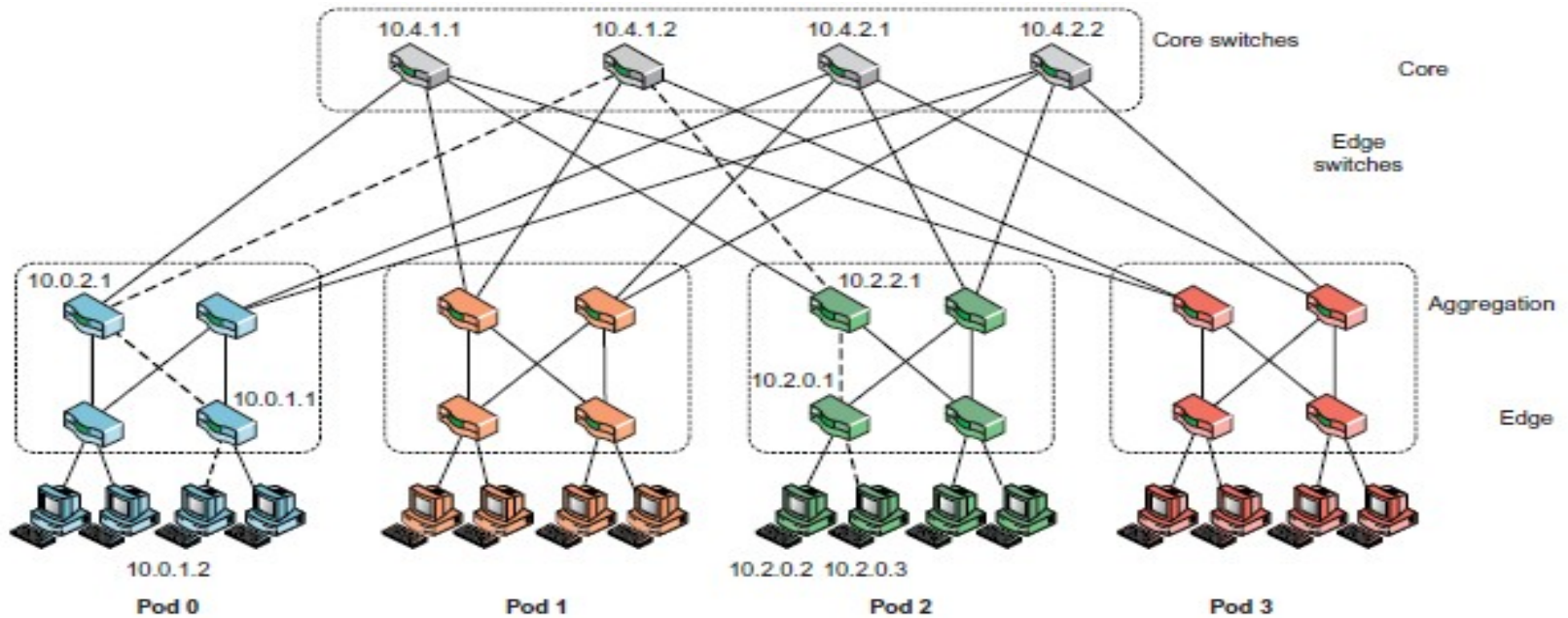


FIGURE 4.10

A fat-tree interconnection topology for scalable data-center construction.

(Courtesy of M. Al-Fares, et al. [2])

Fat-tree switch network design for data-center construction



- The topology is organized into two layers.
 - Server nodes are in the bottom layer,
 - Edge switches are used to connect the nodes in the bottom layer.
 - Aggregation Switches-The upper layer aggregates the lower-layer edge switches.
 - A group of aggregation switches, edge switches, and their leaf nodes form a pod.
 - Core switches provide paths among different pods.

Fat-tree switch network design for data-center construction

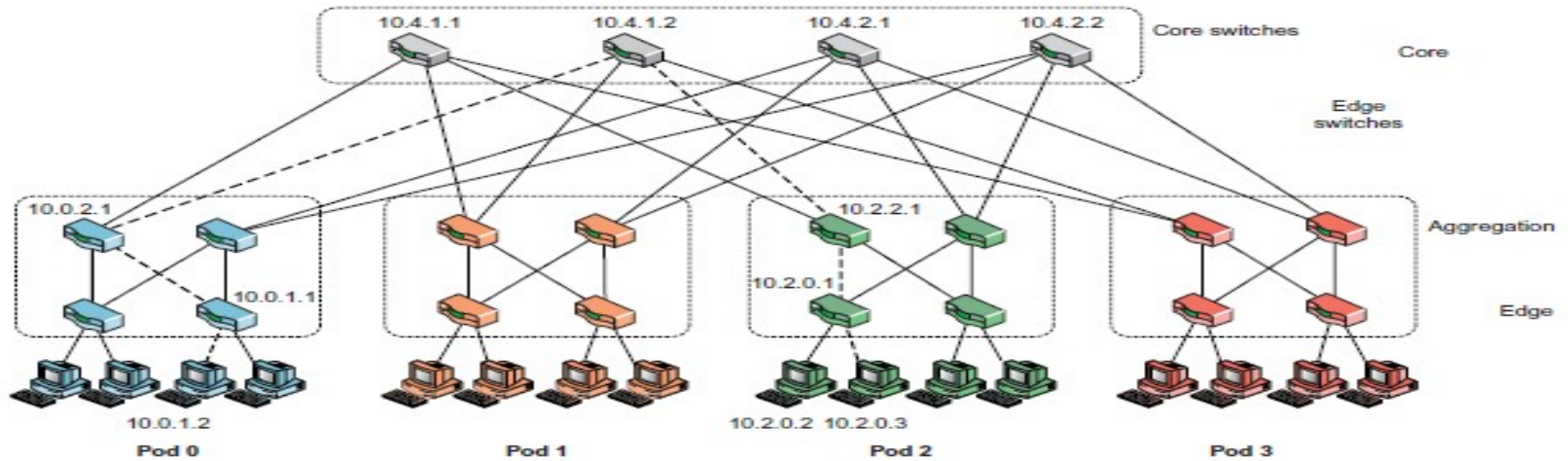


FIGURE 4.10

A fat-tree interconnection topology for scalable data-center construction.

(Courtesy of M. Al-Fares, et al. [2])

- The failure of an aggregation switch will not affect the connectivity of the whole network?
- The failure of an edge switch can only effect a small number of end server nodes
- The extra switches in a pod provide higher bandwidth to support cloud applications

Fat-tree switch network design for data-center construction

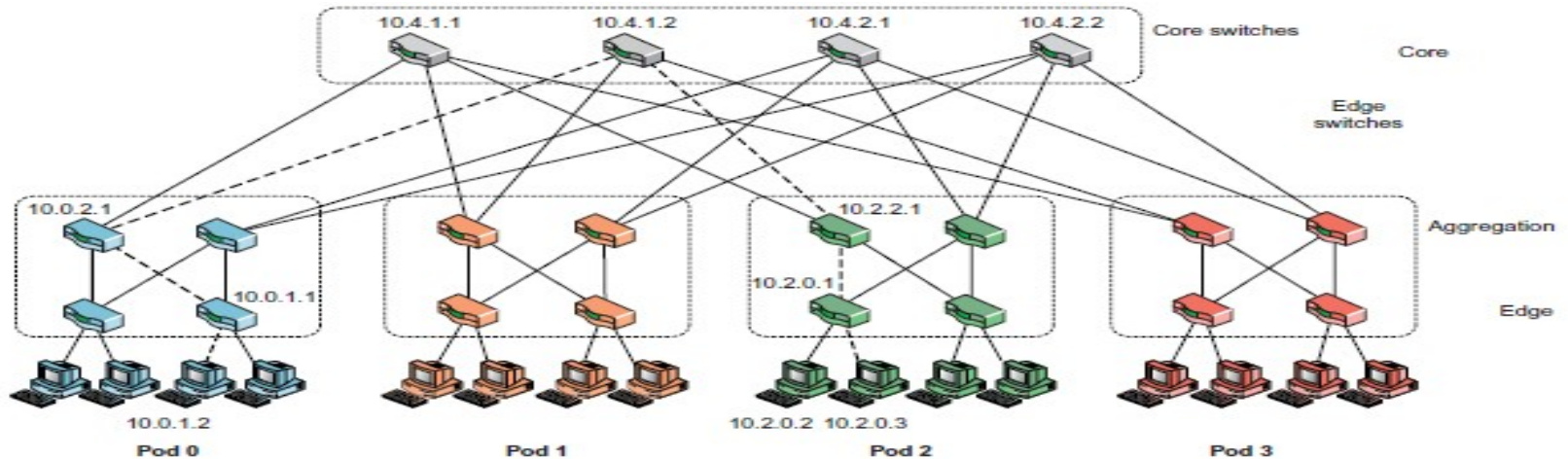


FIGURE 4.10

A fat-tree interconnection topology for scalable data-center construction.

(Courtesy of M. Al-Fares, et al. [2])

- The fat-tree structure provides **multiple paths between any two server nodes**.
- This provides **fault-tolerant capability** with an alternate path in case of some isolated link failures.

Fault Tolerance

- The interconnection network should provide
 - Mechanism to [tolerate link or switch failures](#).
 - **Multiple paths between any two server nodes in a data center**.
 - Fault tolerance of servers is achieved by [replicating data and computing among redundant servers](#).
 - Similar [redundancy](#) should apply to the [network structure](#). Both software and hardware network redundancy apply to cope with potential failures.
 - **The software layer should be aware of network failures. Packet forwarding should avoid using broken links**.
 - **The software drivers for network support should handle this transparently without affecting cloud operations**

Graceful Degradation

- In case of failures,
 - the network structure should degrade gracefully amid limited node failures.
 - Hot-swappable components are desired.
- There should be no critical paths or critical points which may become a single point of failure that pulls down the entire system.

Switch-centric Data-Center Design

- There are two approaches to building data-center-scale networks:
 - Switch centric
 - Server-centric.

Switch-centric Data-Center Design

In a switch-centric network,

- Switches are used to connect the server nodes.
- The switch-centric design does not affect the server side.
- No modifications to the servers are needed.

The server-centric design

- Does modify the operating system running on the servers.
- **Special drivers are designed for relaying the traffic.**
- Switches still have to be organized to achieve the connections.

Modular Data Center in Shipping Containers

- A modern data center is structured as a shipyard of server clusters housed in truck-towed containers.
- The Figure shows the housing of multiple server racks in a truck-towed container in the SGI ICE Cube modular datacenter



Modular Data Center in Shipping Containers

- Portable method of deploying data center capacity.
- A modular data center can be placed anywhere data capacity is needed.





Multi-Story Modular Data Center Design



A 40-foot Portable Modular Data Center.

Modular Data Center in Shipping Containers

Rack server



Blade server



- Inside the container, **hundreds of blade servers are housed in racks surrounding the container walls.**
- An **array of fans forces the heated air** generated by the server racks to go **through a heat exchanger**, which cools the air for the next rack on a continuous loop.

Rack server



- A rack server, or rack-mounted server, is any server that is built specifically to be mounted within a server rack.
- Rack servers are a general-purpose machine that can be configured to support a wide range of requirements.
- So it can be secured into the rack using mounting screws or rails, depending on the design.

Blade server



- A blade server is a modular server that allows multiple servers to be housed in a smaller area.
- These servers are physically thin and typically only have CPUs, memory, integrated network controllers, and sometimes storage drives built in.
- Any video cards or other components that are needed will be facilitated by the server chassis.

Modular Data Center in Shipping Containers

- The SGI ICE Cube container can house 46,080 processing cores or 30 PB of storage per container.
- Large-scale data center built with modular containers appear as a big shipping yard of container trucks.



Motivation?

- This container-based data center was motivated by
 - demand for lower power consumption,
 - higher computer density, and
 - mobility to relocate data centers to better locations
 - with lower electricity costs,
 - better cooling water supplies, and
 - cheaper housing for maintenance engineers.

Motivation?

- Data centers usually are built at a site where leases and utilities for electricity are cheaper and cooling is more efficient.
- In addition to location selection and power savings, one must consider server monitoring and security management in data centers.
- MDC with a high degree of mobility to instead of being fixed at a location

Container Data-Center Construction

- The data-center module is housed in [a truck-towable container.](#)
- The modular container design includes the [network, computer, storage, and cooling gear.](#)
- One needs to increase cooling efficiency by varying the water and airflow with better airflow management. Another concern is to meet [seasonal load requirements.](#)

Container Data-Center Construction

- The construction of a container-based data center
 - may start with one system (server), then
 - move to a rack system design, and
 - finally to a container system.
 - This staged development may take different amounts of time and demand increasing costs.
 - Building a rack of 40 servers may take a half a day.
 - Extending this to a whole container system with multiple racks for 1000 servers requires layout of the floor space with power, networking, cooling and complete testing.

Container Data-Center Construction

- The container must be designed to be weatherproof and easy to transport.
- Modular data-center construction and testing may take a few days to complete if all components are available and power and water supplies are handy.

Portability

- The modular data-center approach supports many cloud service applications.
- For example, the health care industry will benefit by installing a data center at all clinic sites.

Interconnection of Modular Data Centers

- Container based Data Center modules are meant for construction of even larger data centers using a farm of container modules

A Server-Centric BCube Network for a Modular Data Center

Level 1

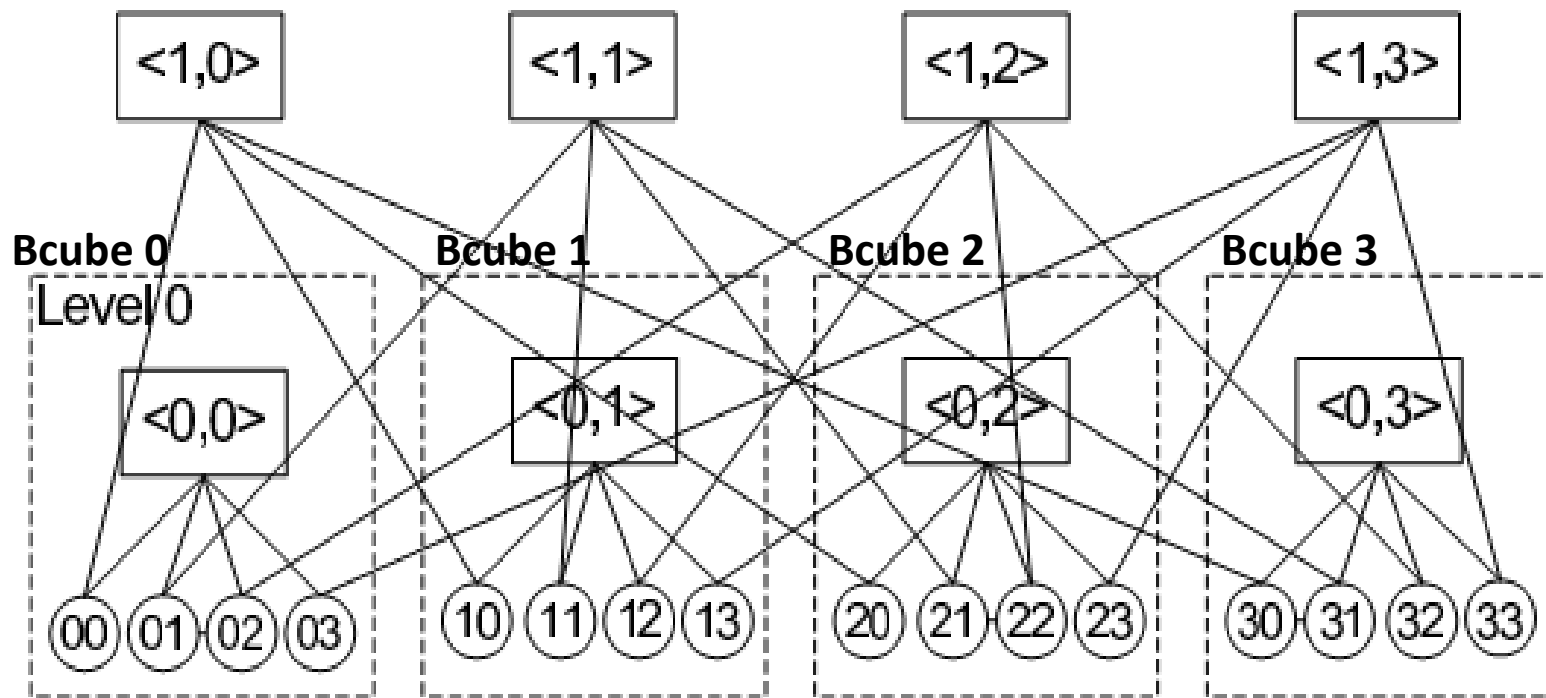
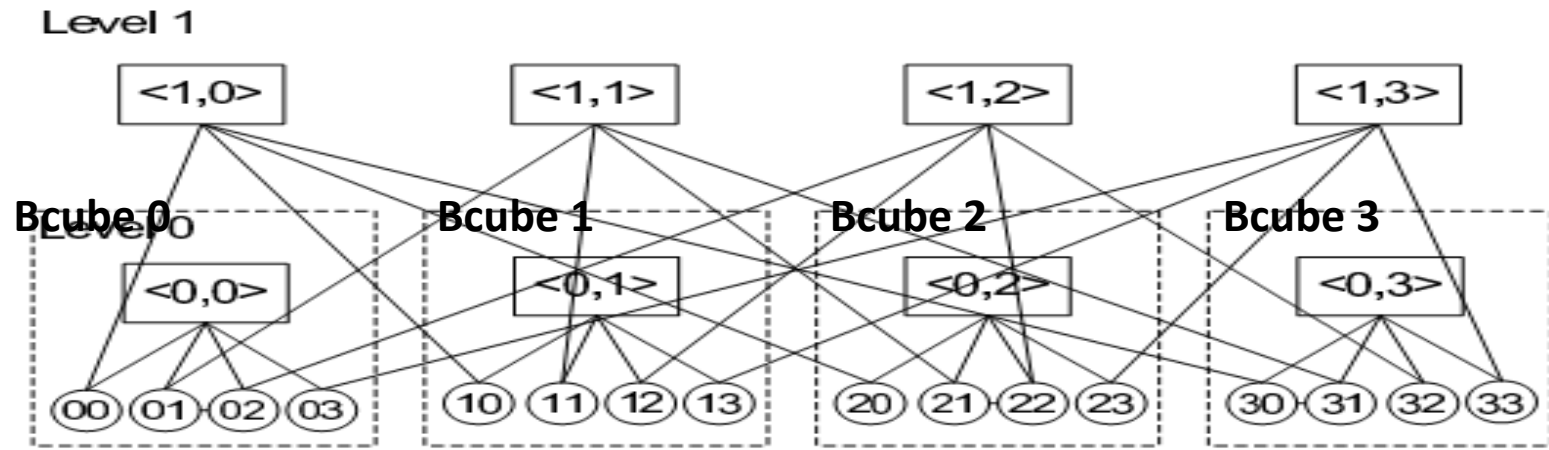


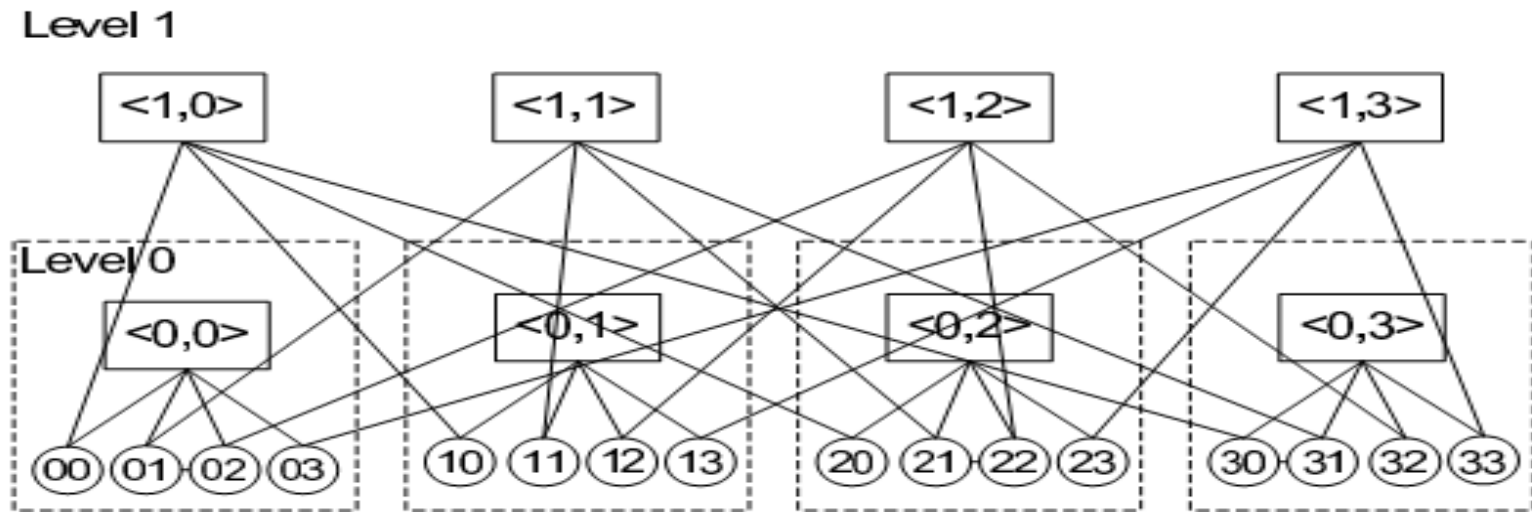
Figure BCube: A High Performance, server-centric network for modular datacenters. (Courtesy of C. Guo, et al, *ACM SIGCOMM Computer Communication Review*, Oct. 2009. [25]).

A Server-Centric BCube Network for a Modular Data Center



- The servers are represented by circles, and switches by rectangles.
- The BCube provides a layered structure.
- The bottom layer contains all the server nodes and they form Level 0.
- Level 1 switches form the top layer of BCube₀.

A Server-Centric BCube Network for a Modular Data Center

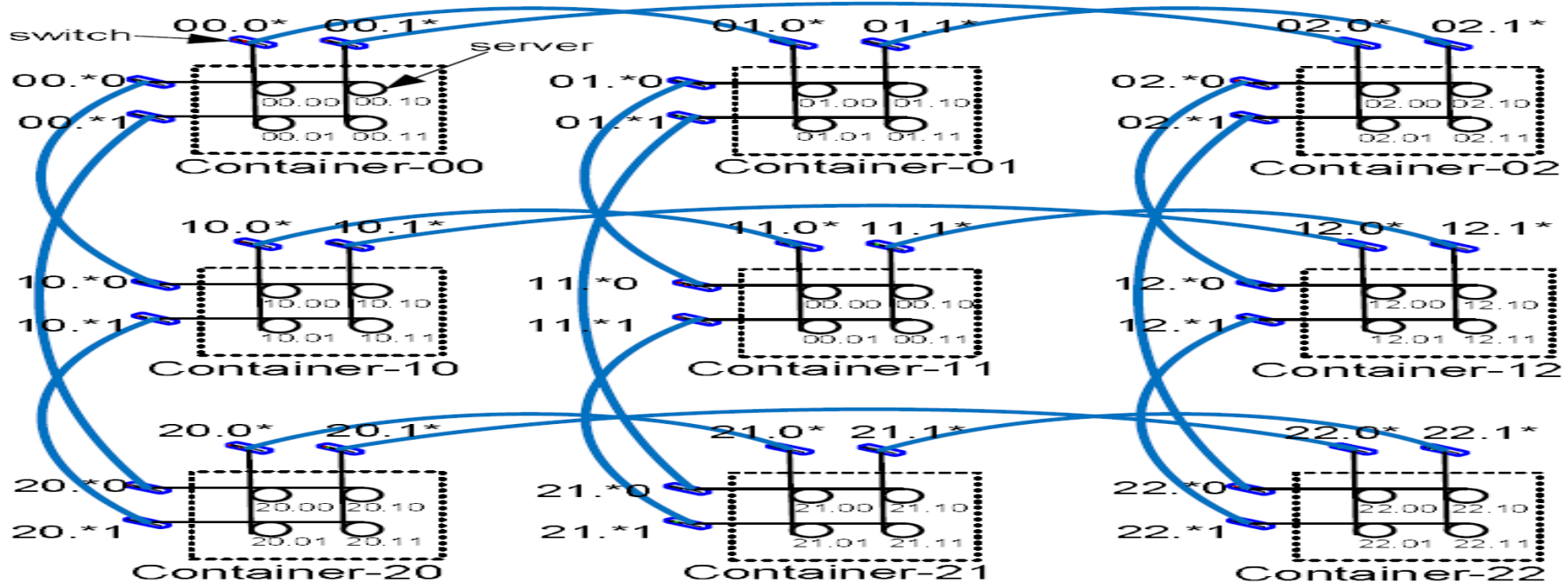


- The BCube provides multiple paths between any two nodes.
- Multiple paths provide extra bandwidth to support communication patterns in different cloud applications.
- The Bcube provides a kernel module in the server OS to perform routing operations.

Inter-Module Connection Networks

- The BCube is commonly used **inside a server container**.
- The containers are considered the building blocks for data centers.
- Thus, despite the design of the inner container network, **one needs another level of networking among multiple containers**.

Inter-Module Connection Networks



- A network topology for intercontainer connection using BCube network as building blocks.
- The proposed network was named MDCube (for Modularized Datacenter Cube).
- This network connects multiple BCube containers by using high-speed switches in the BCube.

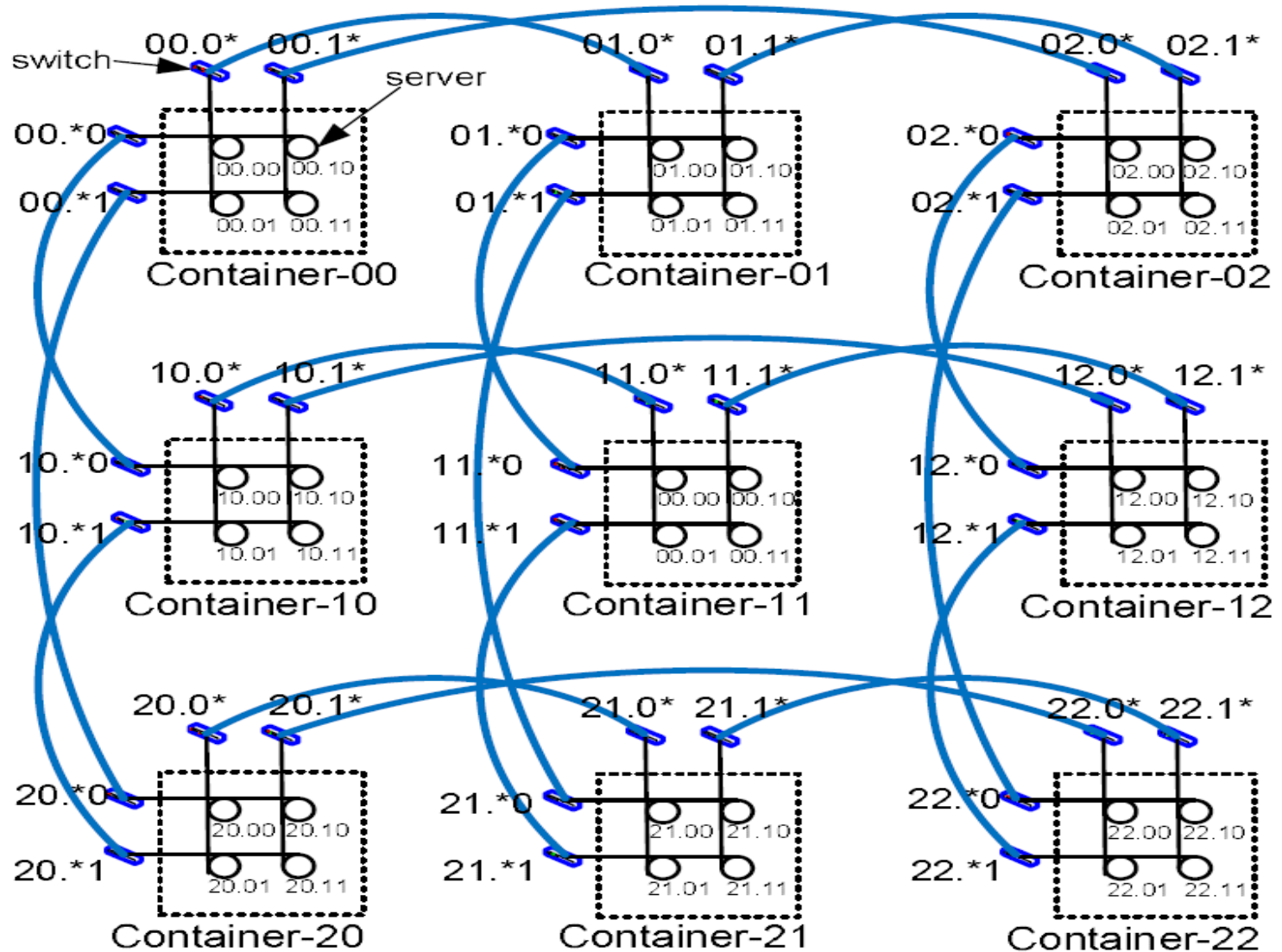


Figure 4.13 A 2-D MDCube is constructed from 9 BCube containers.

Data-Center Management Issues

Here are basic requirements for managing the resources of a data center.

- **Making common users happy**
 - The data center should be designed to provide quality service to the majority of users for at least 30 years.
- **Controlled information flow**
 - Information flow should be streamlined. Sustained services and high availability (HA) are the primary goals.
- **Multiuser manageability –**
 - The system must be managed to support all functions of a data center, including traffic flow, database updating, and server maintenance.
- **Scalability to prepare for database growth –**
 - The system should allow growth as workload increases. The storage, processing, I/O, power, and cooling subsystems should be scalable.

Data-Center Management Issues

- **Reliability** in virtualized infrastructure
 - Failover, fault tolerance, and VM live migration should be integrated to enable recovery of critical applications from failures or disasters.
- **Low cost to both users and providers** –
 - The cost to users and providers of the cloud system built over the data centers should be reduced, including all operational costs.
- **Security enforcement and data protection**-
 - Data privacy and security defense mechanisms must be deployed to protect the data center against network attacks and system interrupts and to maintain data integrity from user abuses or network attacks.
- **Green information technology** –
 - Saving power consumption and upgrading energy efficiency are in high demand when designing and operating current and future data centers.